

## Block diagrams and Signal flow graphs:

### Transfer functions and Block diagram algebra

The input-output behaviour of a linear system or element of a linear system is given by its transfer function,

$$G(s) = \frac{C(s)}{R(s)}$$

where  $R(s)$  = Laplace transform of the input variable;  
and  $C(s)$  = Laplace transform of the output variable.

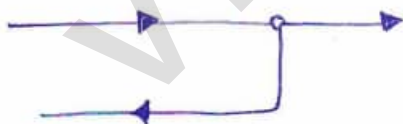
A convenient graphical representation of this behaviour is the block diagram as shown below.



The summing or differencing of signals is indicated by

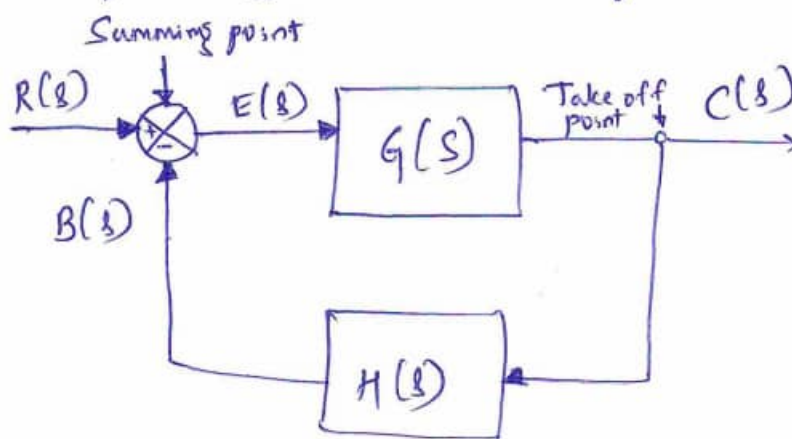


The take-off point of a signal is represented by,



### Block diagram of a closed-loop system:

The following figure shows the block diagram of a negative feedback system.



(2)

With reference to the figure, the terminology used in block diagrams of control systems is given below.

$R(s)$  = Reference input,  $C(s)$  = Output signal

$B(s)$  = Feedback signal,  $E(s)$  = Actuating signal

$G(s) = \frac{C(s)}{E(s)}$  = forward path transfer function

$H(s)$  = Transfer function of the feedback elements

$G(s)H(s)$  = Loop transfer function.

$T(s) = C(s)/R(s)$  = Closed-loop transfer function.

From the above figure we have,

$$G(s) = \frac{C(s)}{E(s)}$$

$$\text{and } C(s) = G(s) \cdot E(s)$$

$$\text{and } E(s) = R(s) - B(s)$$

$$E(s) = R(s) - H(s) \cdot C(s)$$

$$\begin{aligned} \therefore C(s) &= G(s) [R(s) - H(s) \cdot C(s)] \\ &= G(s) \cdot R(s) - H(s) \cdot C(s) \cdot G(s) \end{aligned}$$

$$C(s) + H(s)C(s)G(s) = G(s) \cdot R(s)$$

$$C(s) [1 + G(s)H(s)] = G(s) \cdot R(s)$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)}} = T(s)$$

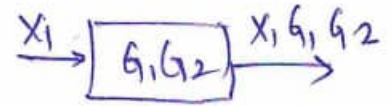
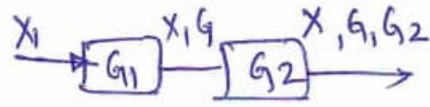
# Rules of Block diagram algebra:

Rule

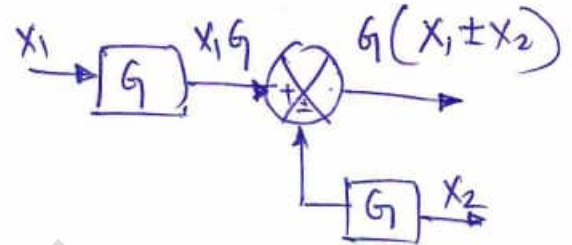
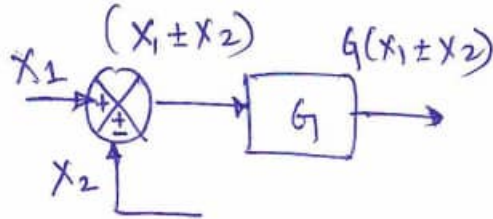
Original diagram

Equivalent diagram

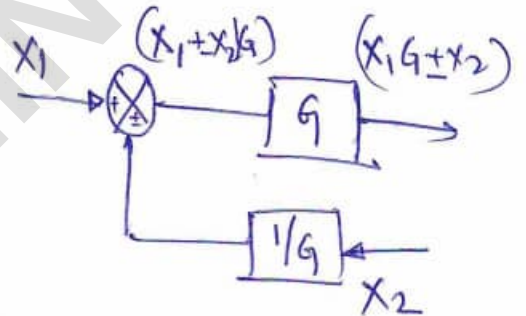
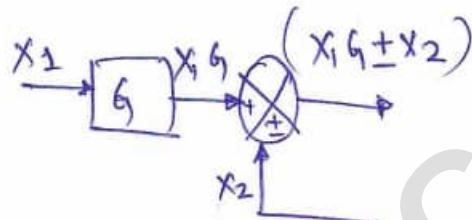
- ① Combining blocks in cascade



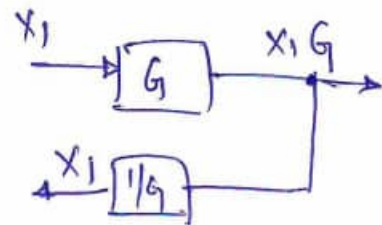
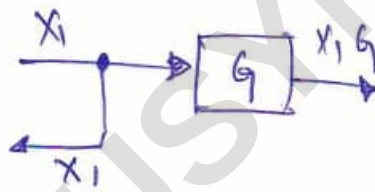
- ② Moving a summing point after the block



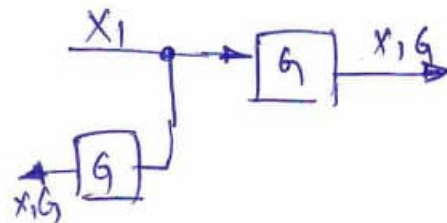
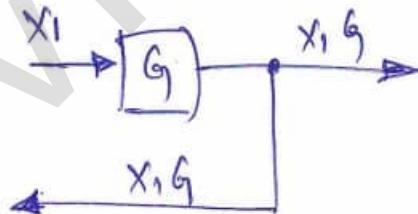
- ③ Moving a summing point ahead of a block



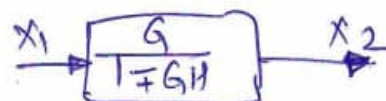
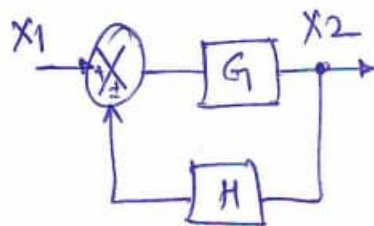
- ④ Moving a take off point after blocks



- ⑤ Moving a take off point ahead of a block

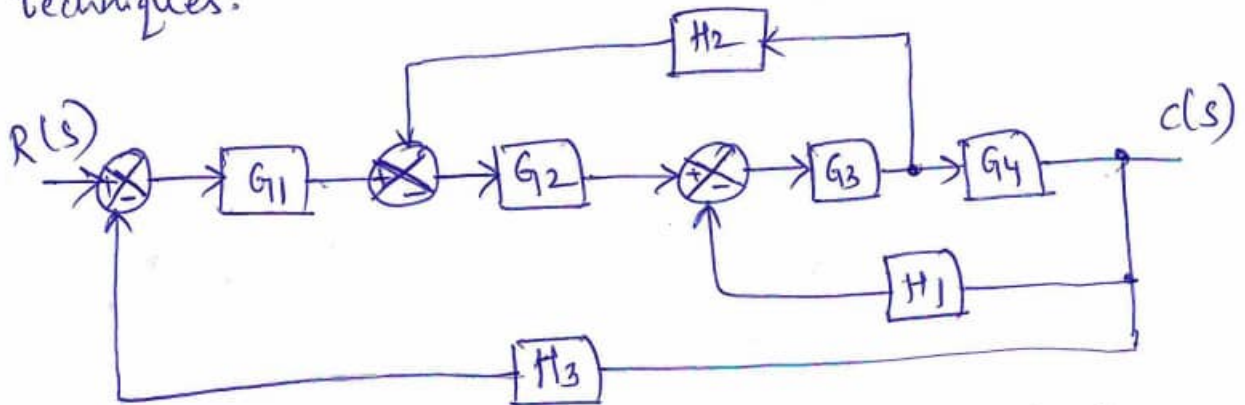


- ⑥ Eliminating a feedback loop

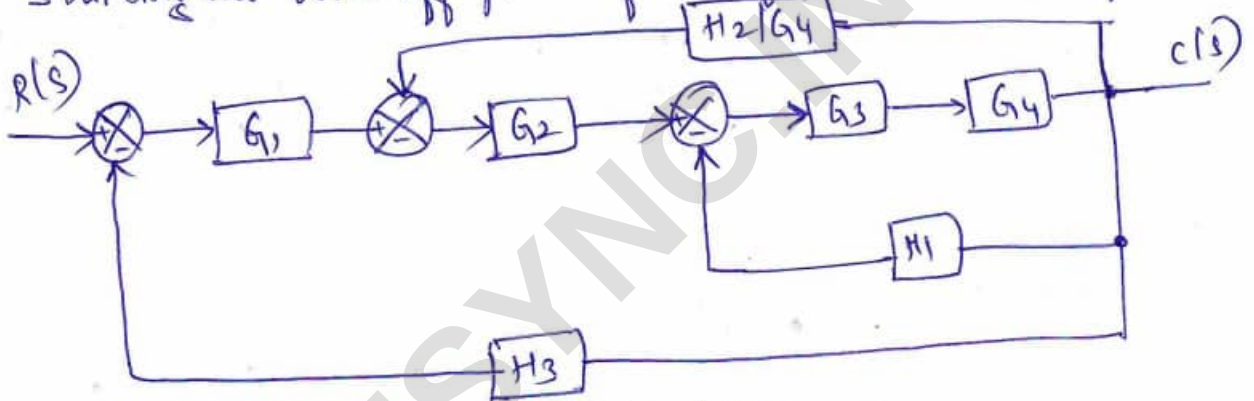




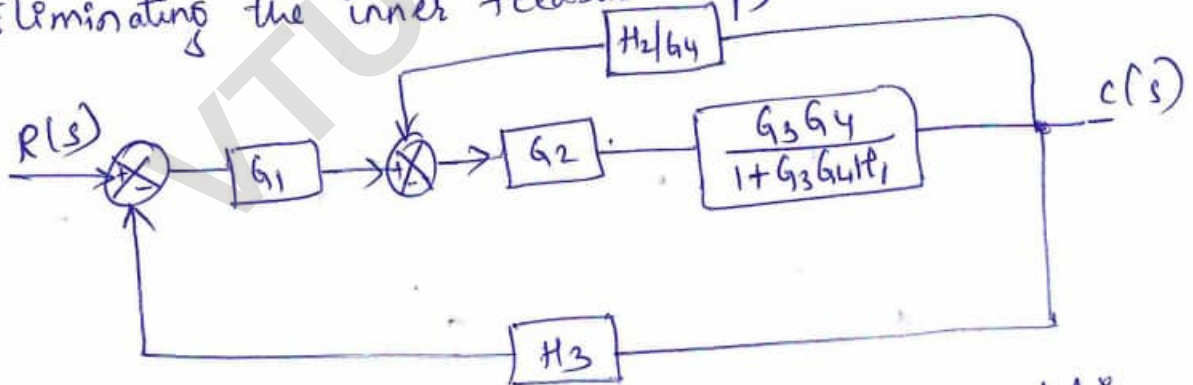
- ① Obtain the transfer function  $T(s) = \frac{C(s)}{R(s)}$  of the following block diagram by making use of reduction techniques.



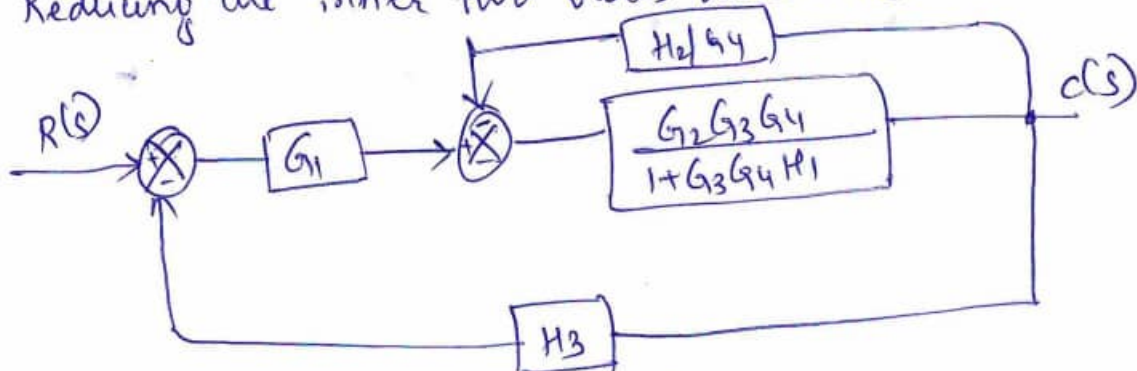
Shifting the take off point of  $H_2$  after the block  $G_4$



Eliminating the inner feedback loop,

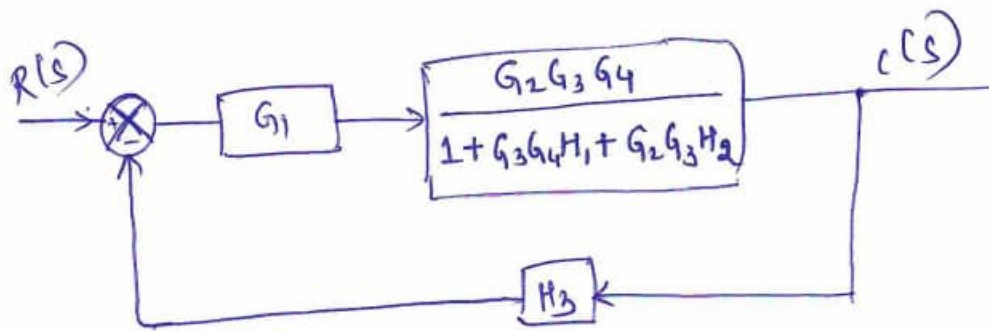


Reducing the inner two blocks which are connected in series,



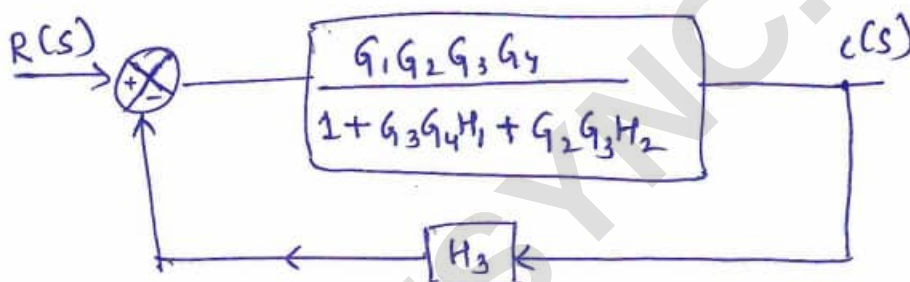
Eliminating the inner feedback loop,

(5)



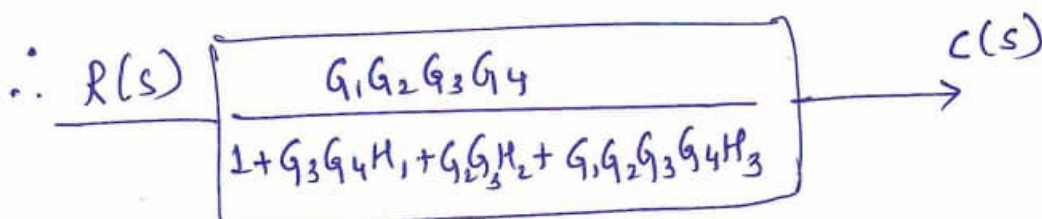
Simplification is as follows,

$$\begin{aligned}
 &= \frac{\frac{G_2 G_3 G_4}{1 + G_3 G_4 H_1}}{1 + \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_1} \cdot \frac{H_2}{G_4}} \Rightarrow \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_1} \times \frac{1 + \cancel{G_3 G_4 H_1}}{1 + G_3 G_4 H_1 + G_2 G_3 H_2} \\
 &\Rightarrow \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2}
 \end{aligned}$$



Finally, eliminating the feedback loop,

$$\begin{aligned}
 &= \frac{\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2}}{1 + \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2} \cdot H_3} \Rightarrow \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2} \cdot \frac{1 + \cancel{G_3 G_4 H_1} + \cancel{G_2 G_3 H_2}}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4 H_3} \\
 &\Rightarrow \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4 H_3}
 \end{aligned}$$



## Procedure to solve Block diagram reduction problems

Step 1: Reduce the blocks connected in series.

Step 2: Reduce the blocks connected in parallel.

Step 3: Reduce the minor internal feedback loops.

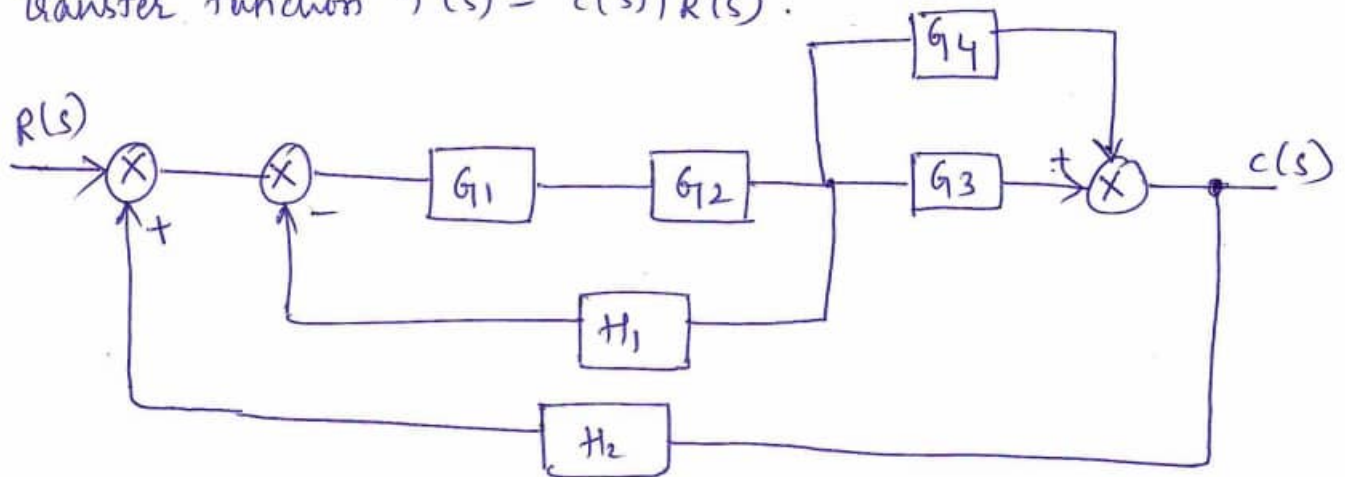
Step 4: As far as possible try to shift take off points towards right and summing points to the left.

Step 5: Repeat steps 1 to 4 till simple form is obtained.

Step 6: Using standard Transfer function of simple closed loop system, obtain the closed loop Transfer function.  $C(s)/R(s)$  of the overall system.

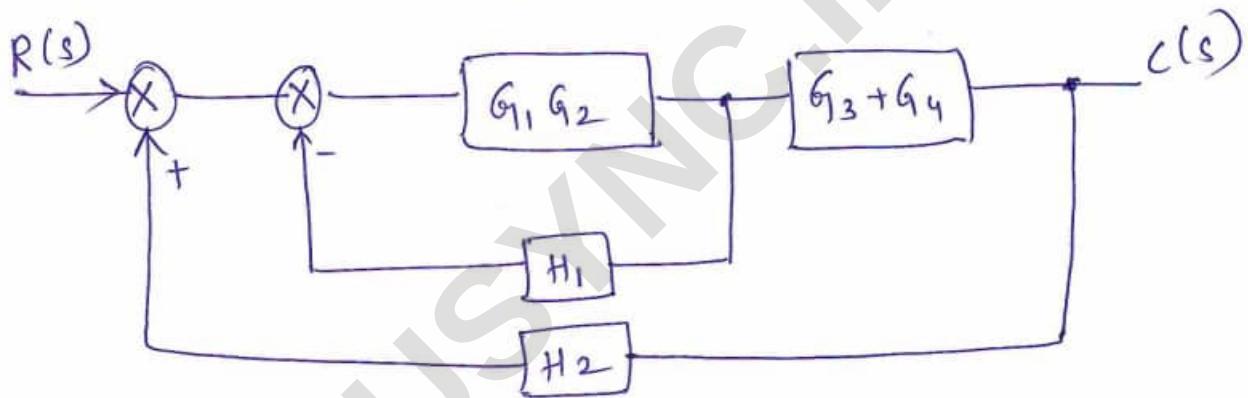


- ② Reduce the following block diagram and obtain the transfer function  $T(s) = C(s)/R(s)$ .

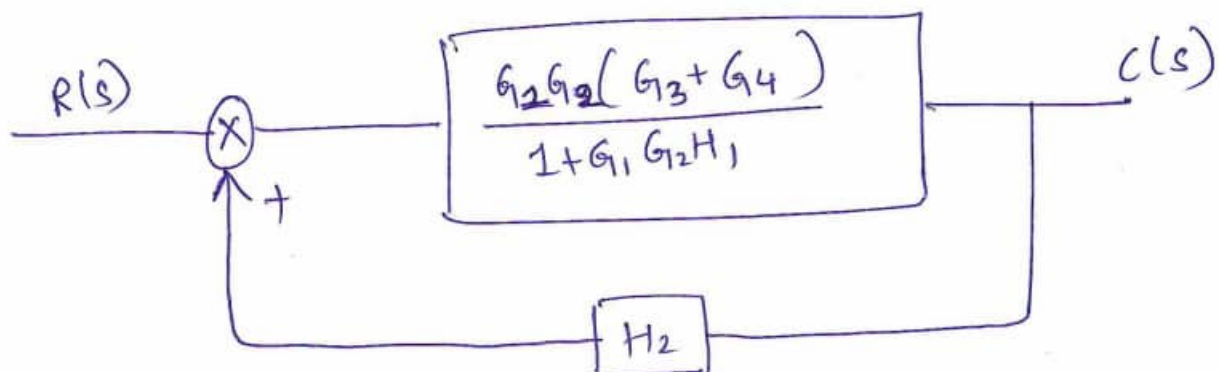
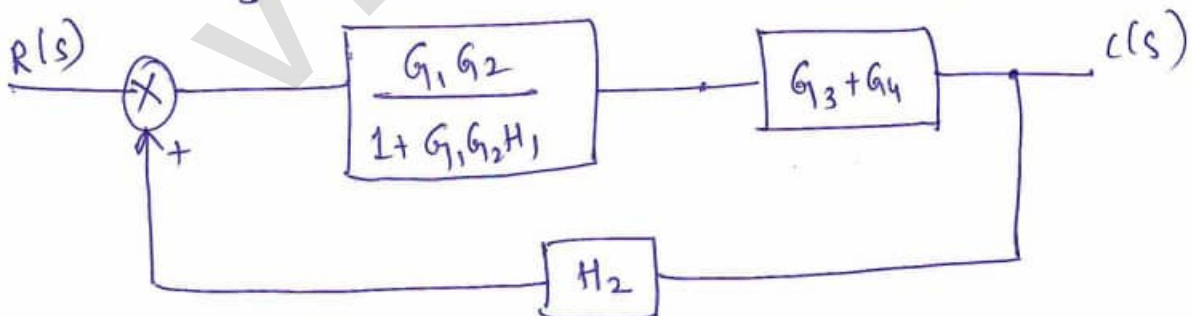


Sol<sup>n</sup>:

$G_1$  and  $G_2$  are in series &  $G_3$  and  $G_4$  are in parallel.



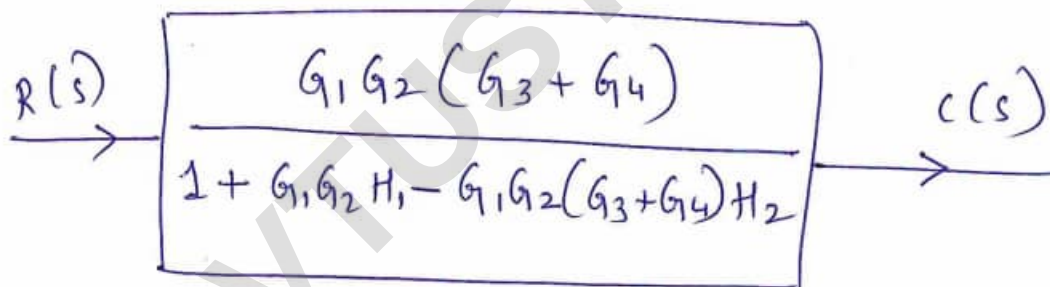
Reducing the inner minor feedback loop:



Finally reducing the feedback loop,

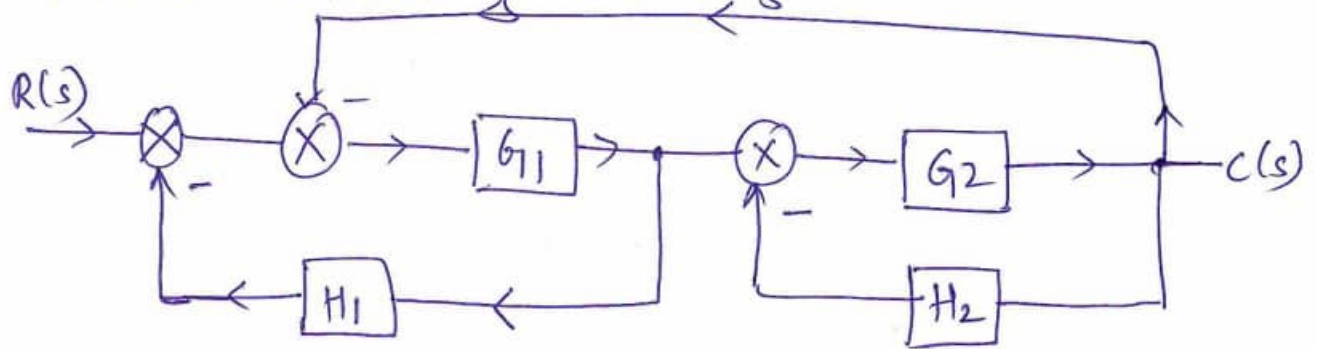
$$\begin{aligned}
 & \frac{G_1 G_2 (G_3 + G_4)}{1 + G_1 G_2 H_1} \\
 = & \frac{1 - \frac{G_1 G_2 (G_3 + G_4) H_2}{1 + G_1 G_2 H_1}}{1 + G_1 G_2 H_1} \\
 = & \frac{G_1 G_2 (G_3 + G_4)}{1 + G_1 G_2 H_1} \times \frac{1 + G_1 G_2 H_1}{1 + G_1 G_2 H_1 - G_1 G_2 (G_3 + G_4) H_2} \\
 = & \frac{G_1 G_2 (G_3 + G_4)}{1 + G_1 G_2 H_1 - G_1 G_2 (G_3 + G_4) H_2}
 \end{aligned}$$

∴ The reduced the block diagram with  $T(s) = C(s)/R(s)$

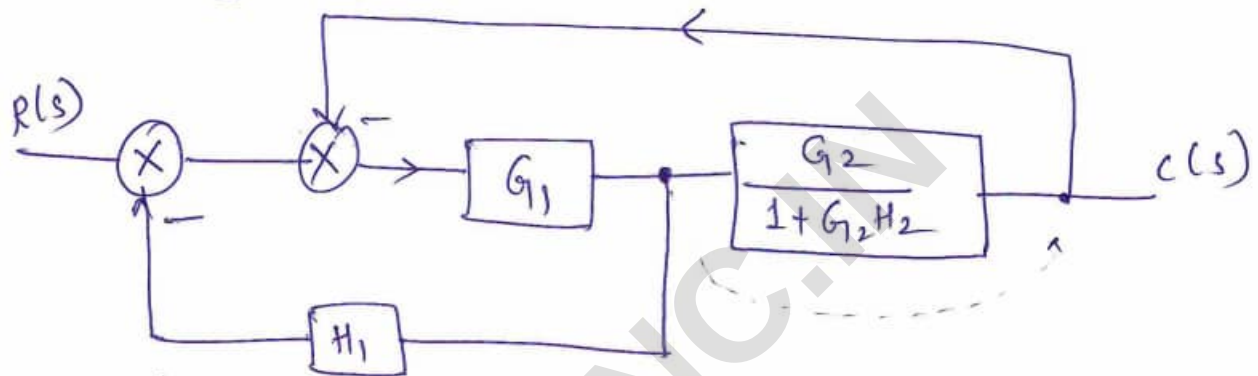




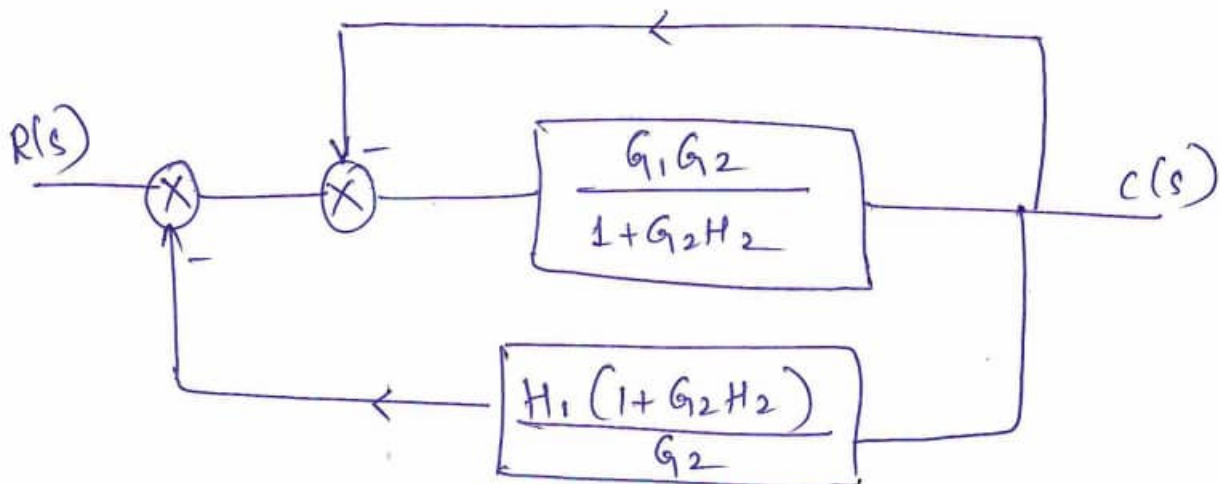
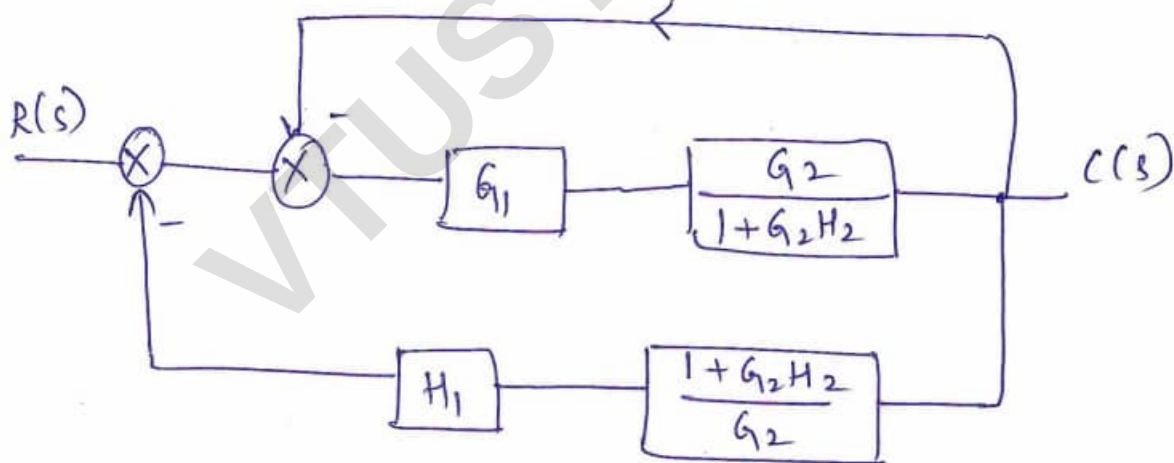
③ Reduce the following block diagram.

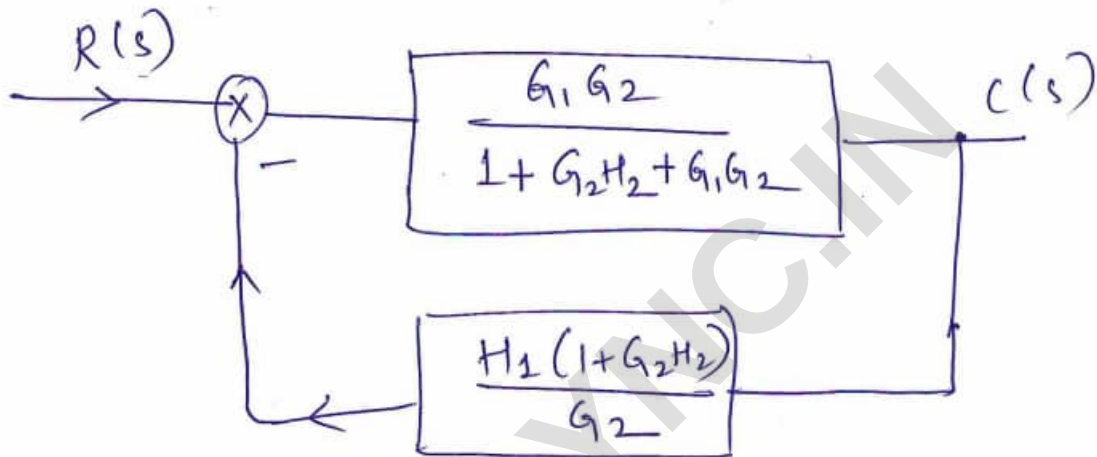
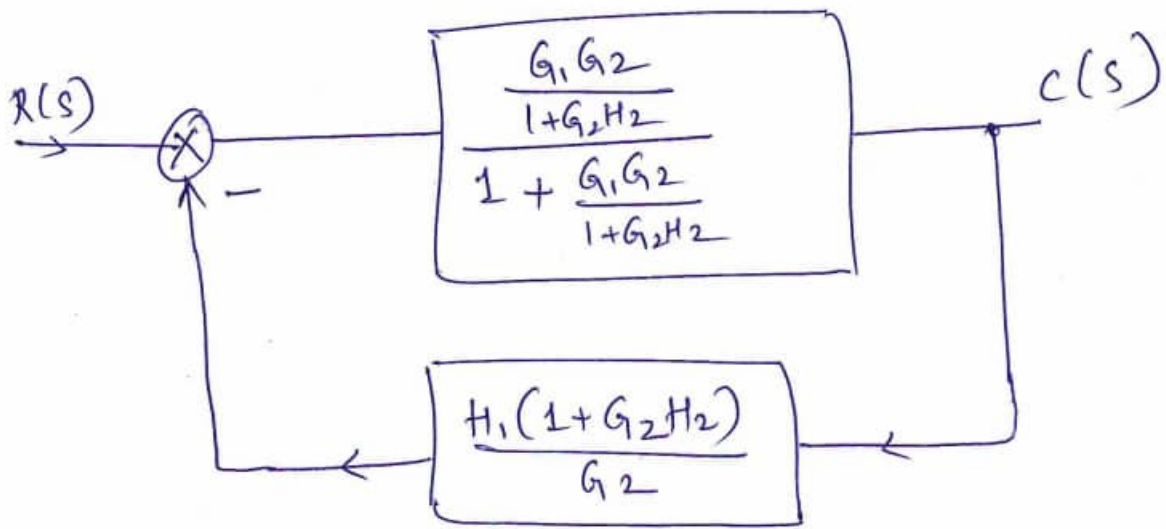


Sol<sup>n</sup>: Reducing the feedback loop with blocks  $G_2$  and  $H_2$



Shifting the take-off point after the block.

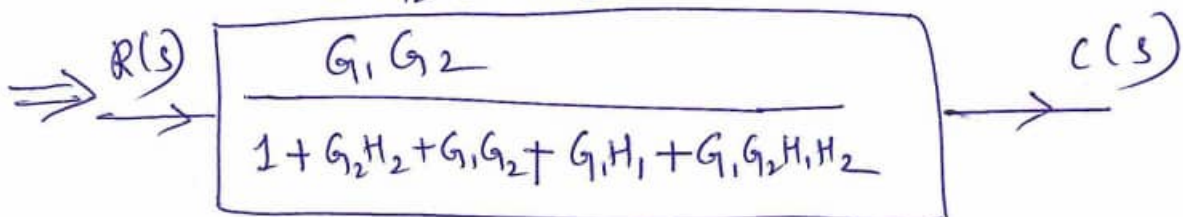




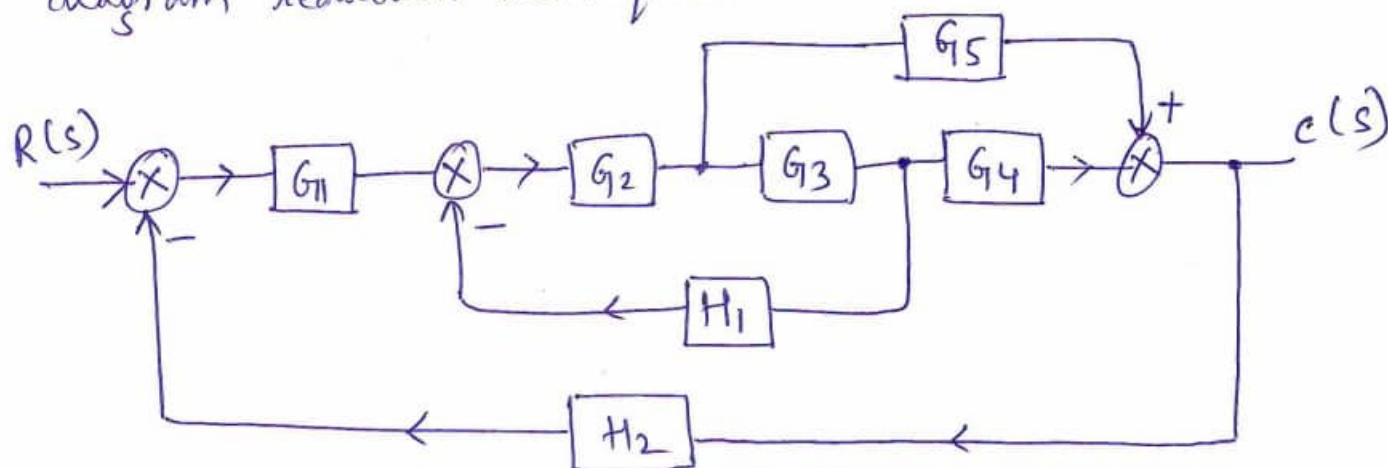
Reducing the feedback loop,

$$= \frac{\frac{G_1 G_2}{1 + G_2 H_2 + G_1 G_2}}{1 + \frac{G_1 G_2}{1 + G_2 H_2 + G_1 G_2} \cdot \frac{H_1 (1 + G_2 H_2)}{G_2}}$$

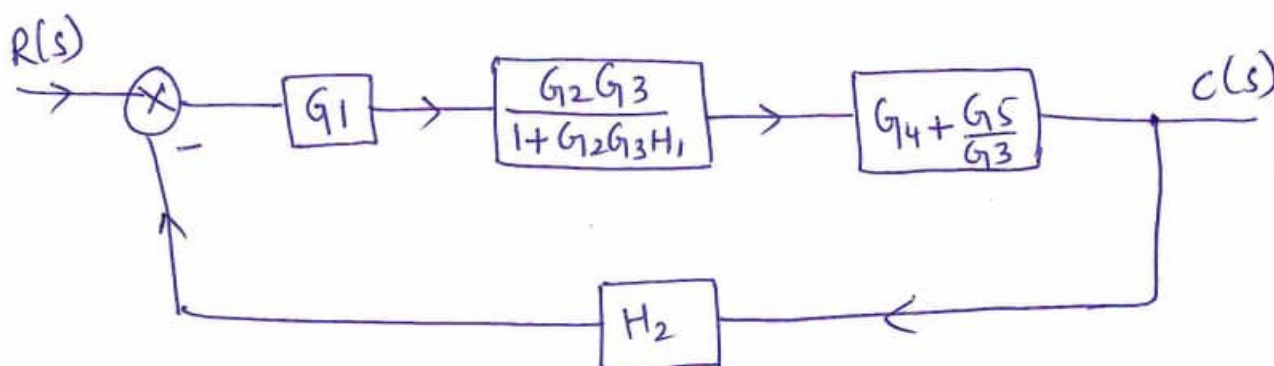
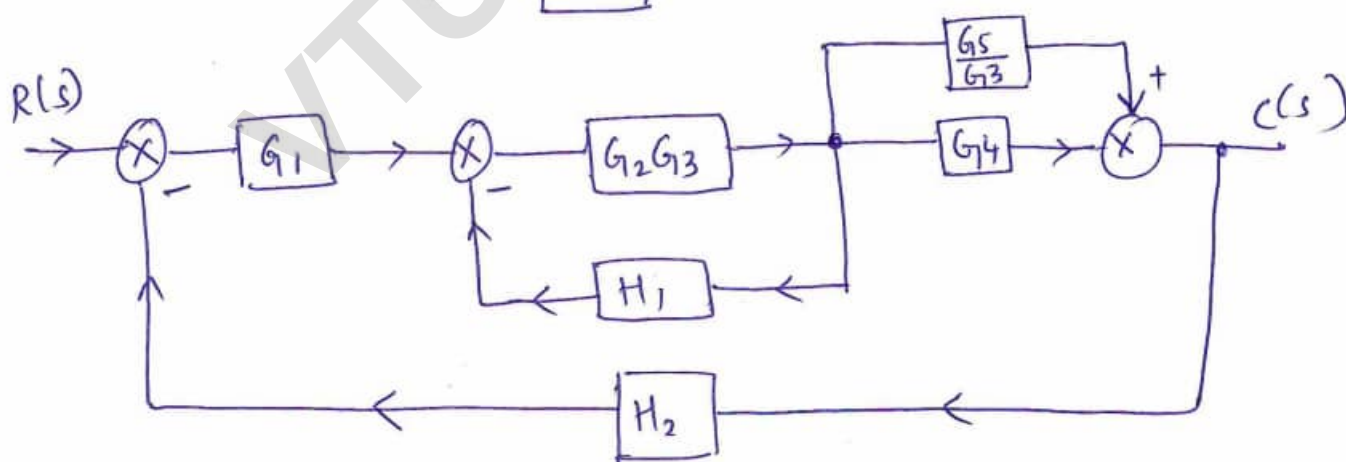
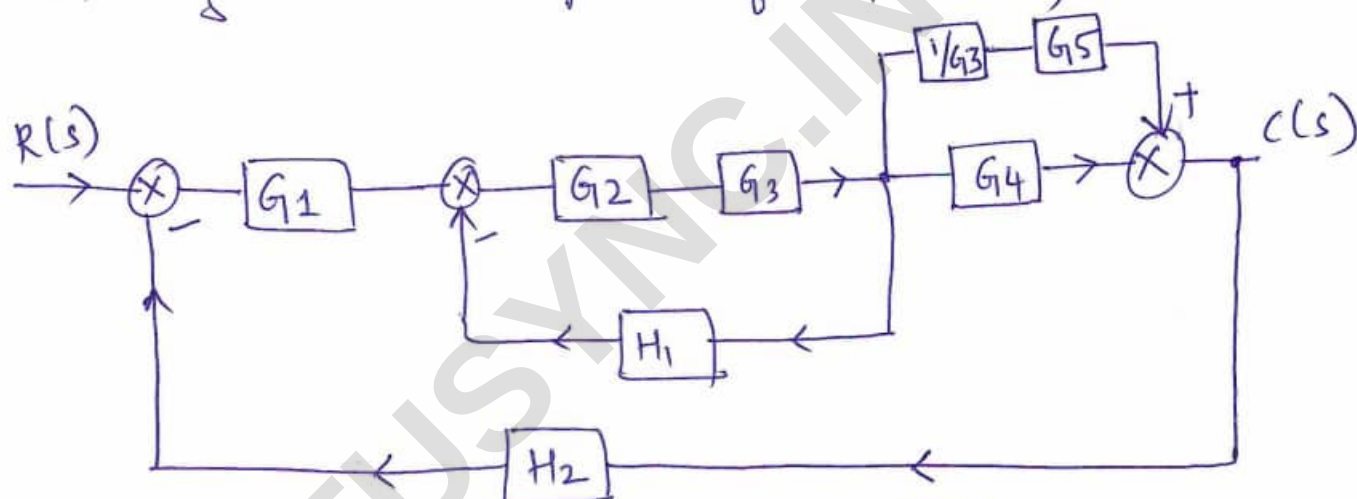
$$= \frac{\frac{G_1 G_2}{1 + G_2 H_2 + G_1 G_2}}{1 + \frac{G_1 H_1 (1 + G_2 H_2)}{1 + G_2 H_2 + G_1 G_2}} \Rightarrow \frac{G_1 G_2}{1 + G_2 H_2 + G_1 G_2} \times \frac{1 + G_2 H_2 + G_1 G_2}{1 + G_2 H_2 + G_1 G_2 + G_1 H_1 (1 + G_2 H_2)}$$



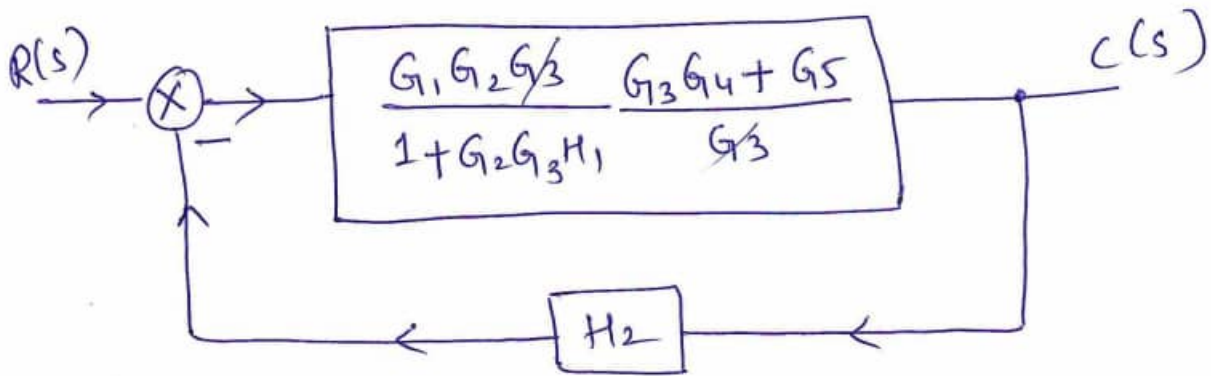
- (4) Reduce the following block diagram using block diagram reduction techniques.



Sol<sup>n</sup>: Shifting the take-off point after  $G_3$  block,

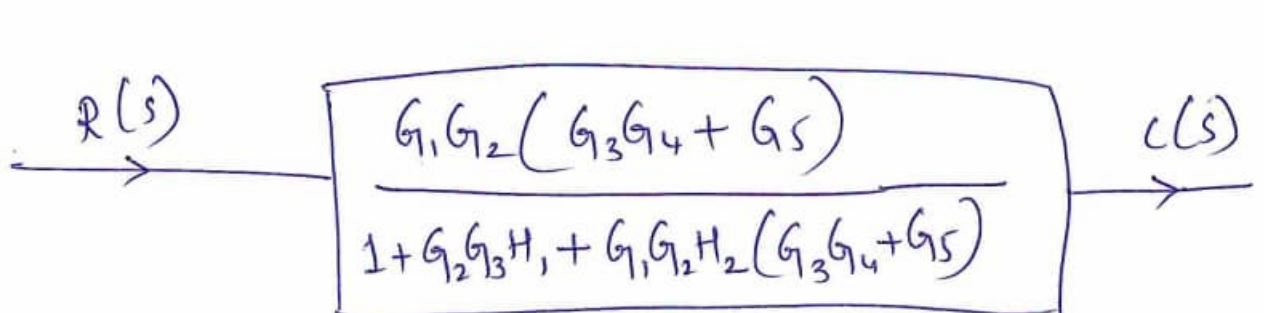




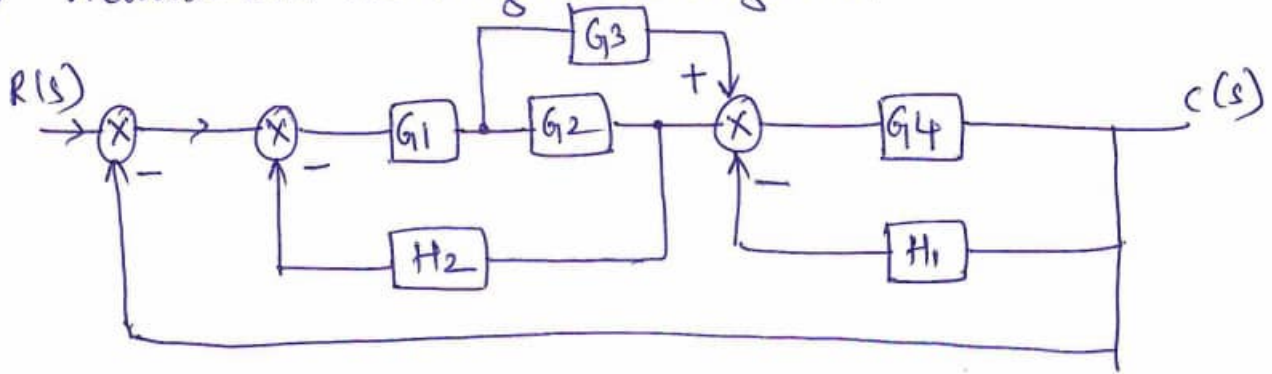


Reducing the feedback loop,

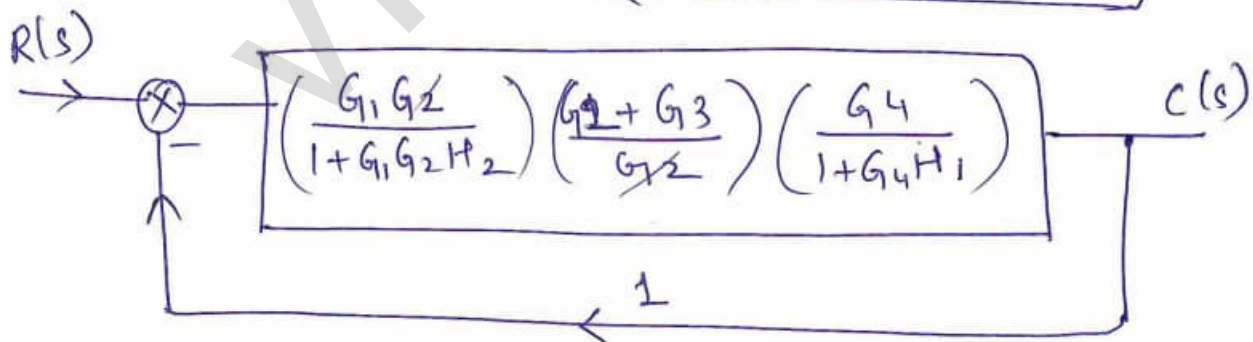
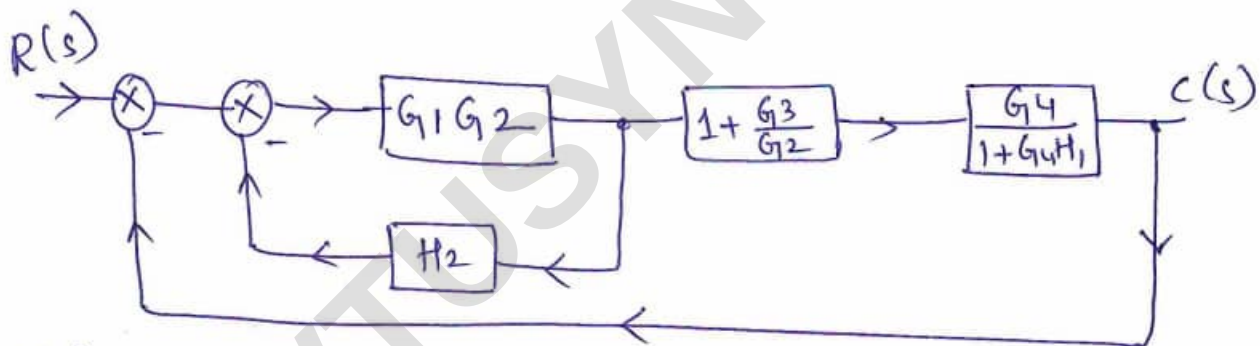
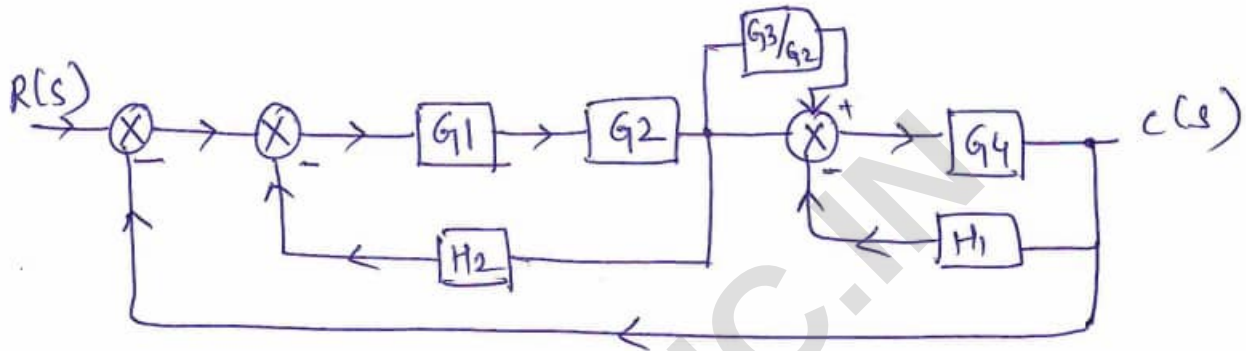
$$\begin{aligned}
 & \frac{G_1 G_2 (G_3 G_4 + G_5)}{1 + G_2 G_3 H_1} \\
 = & \frac{G_1 G_2 (G_3 G_4 + G_5)}{1 + \frac{G_1 G_2 (G_3 G_4 + G_5) H_2}{1 + G_2 G_3 H_1}} \\
 = & \frac{G_1 G_2 (G_3 G_4 + G_5)}{1 + G_2 G_3 H_1} \times \frac{1 + G_2 G_3 H_1}{1 + G_2 G_3 H_1 + G_1 G_2 H_2 (G_3 G_4 + G_5)} \\
 = & \frac{G_1 G_2 (G_3 G_4 + G_5)}{1 + G_2 G_3 H_1 + G_1 G_2 H_2 (G_3 G_4 + G_5)}
 \end{aligned}$$



⑤ Reduce the following block diagram.



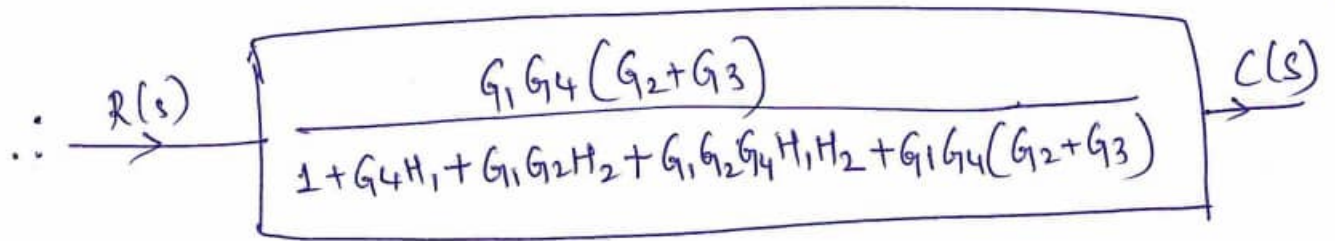
Sol<sup>n</sup>: Shifting the take-off point after  $G_2$ ,



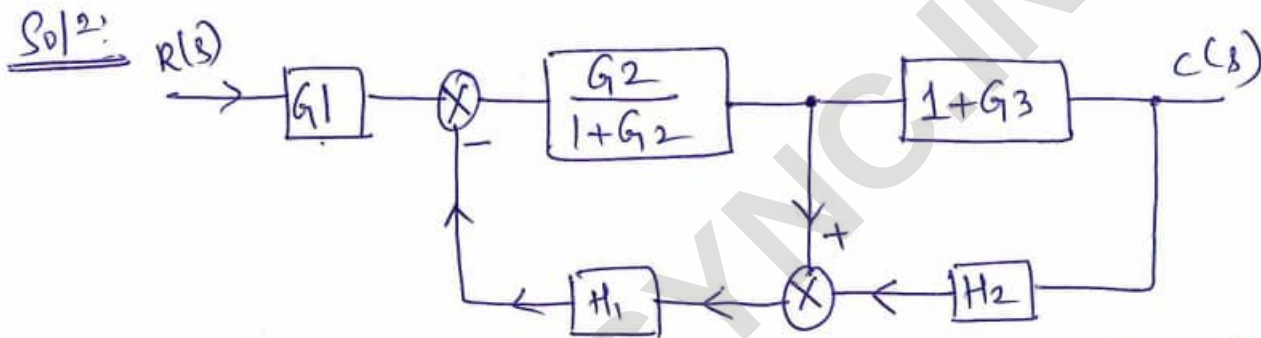
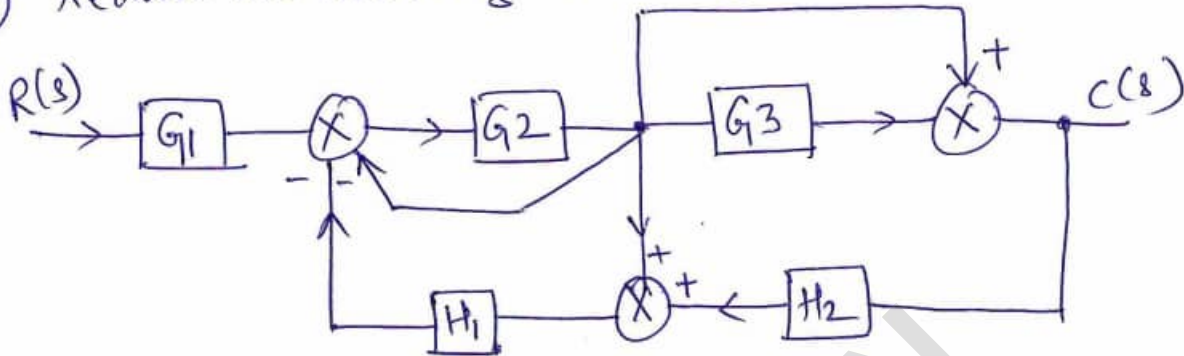
$$= \frac{G_1 G_4 (G_2 + G_3)}{(1 + G_1 G_2 H_2)(1 + G_4 H_1)}$$

$$= \frac{G_1 G_4 (G_2 + G_3)}{1 + \frac{G_1 G_4 (G_2 + G_3)}{(1 + G_1 G_2 H_2)(1 + G_4 H_1)}}$$

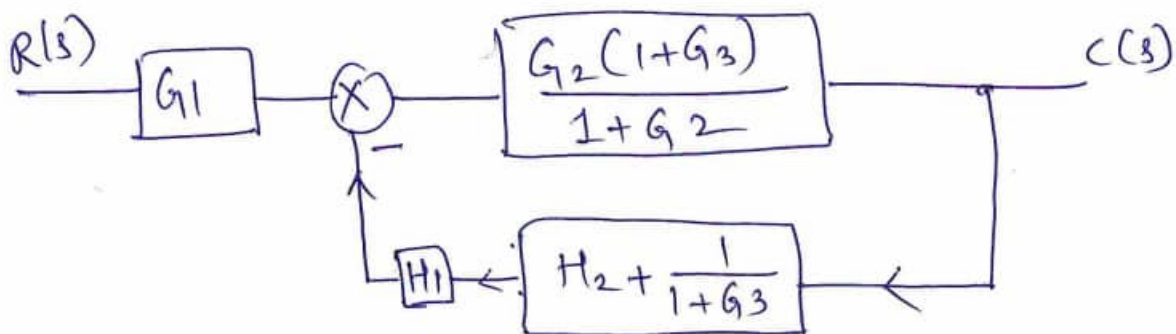
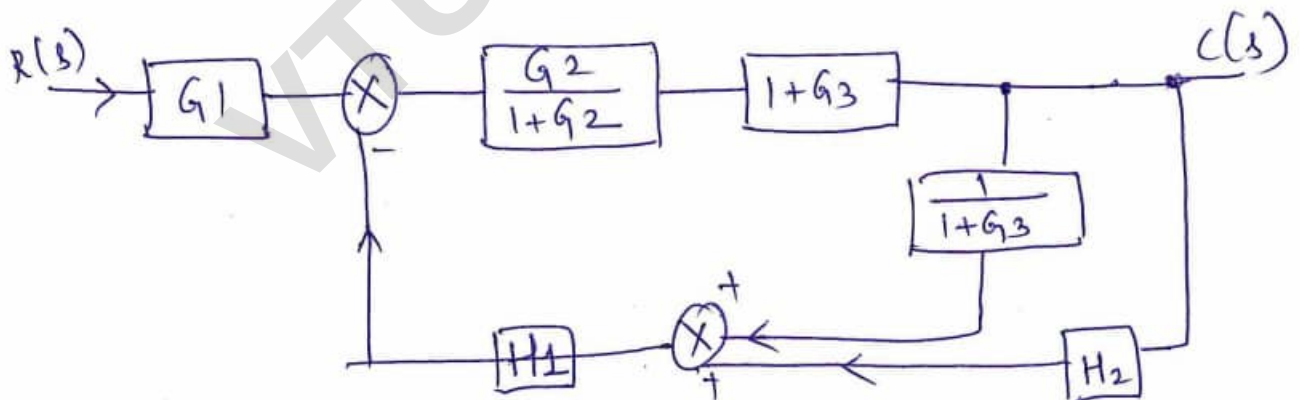
$$= \frac{G_1 G_4 (G_2 + G_3)}{(1 + G_1 G_2 H_2)(1 + G_4 H_1) + G_1 G_4 (G_2 + G_3)}$$



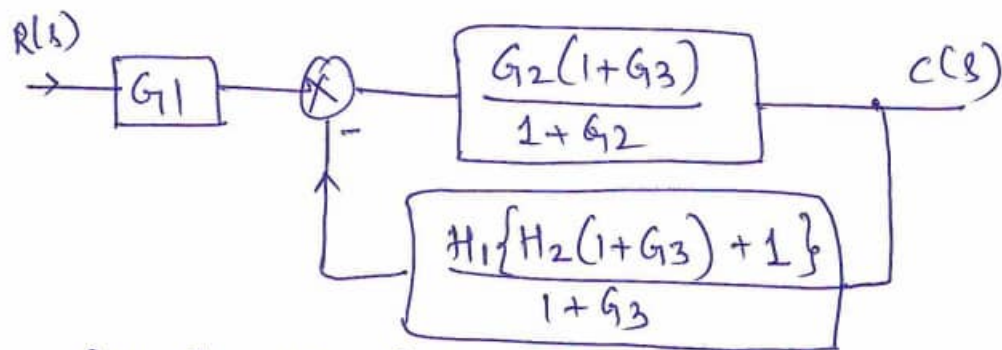
⑥ Reduce the following block diagrams.



Shifting the take-off point after the block  $(1+G_3)$

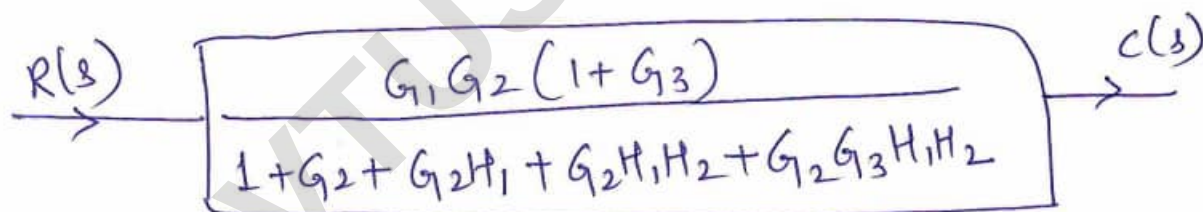
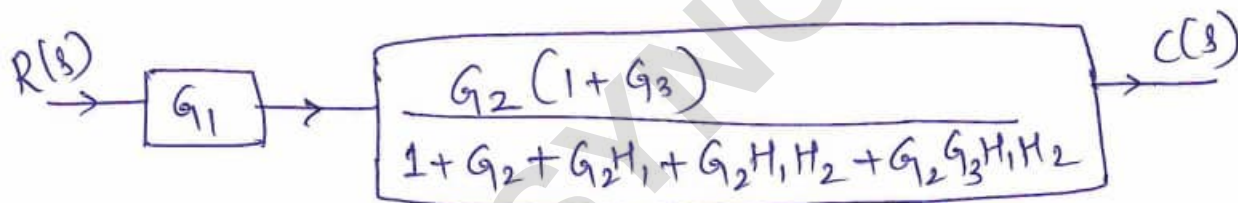




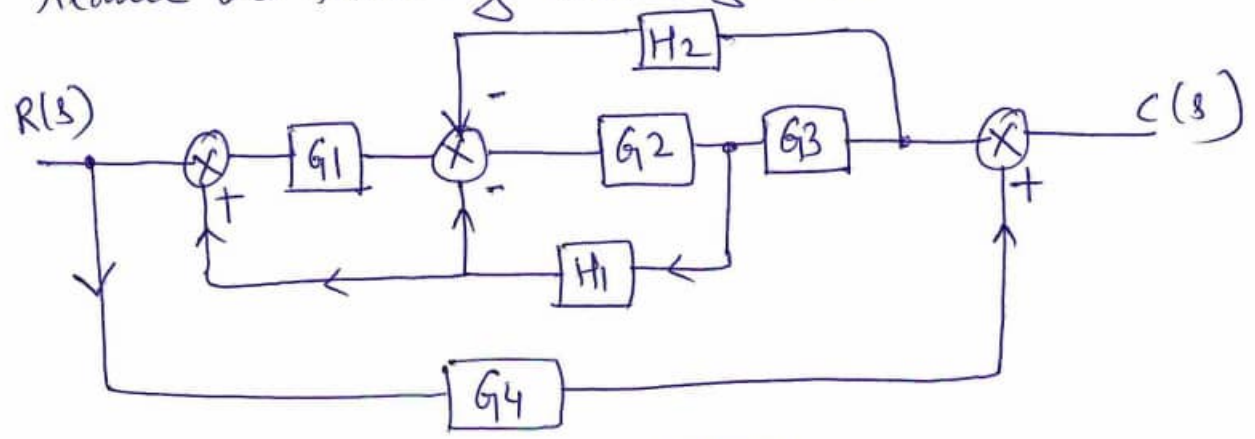


Reducing the feedback loop,

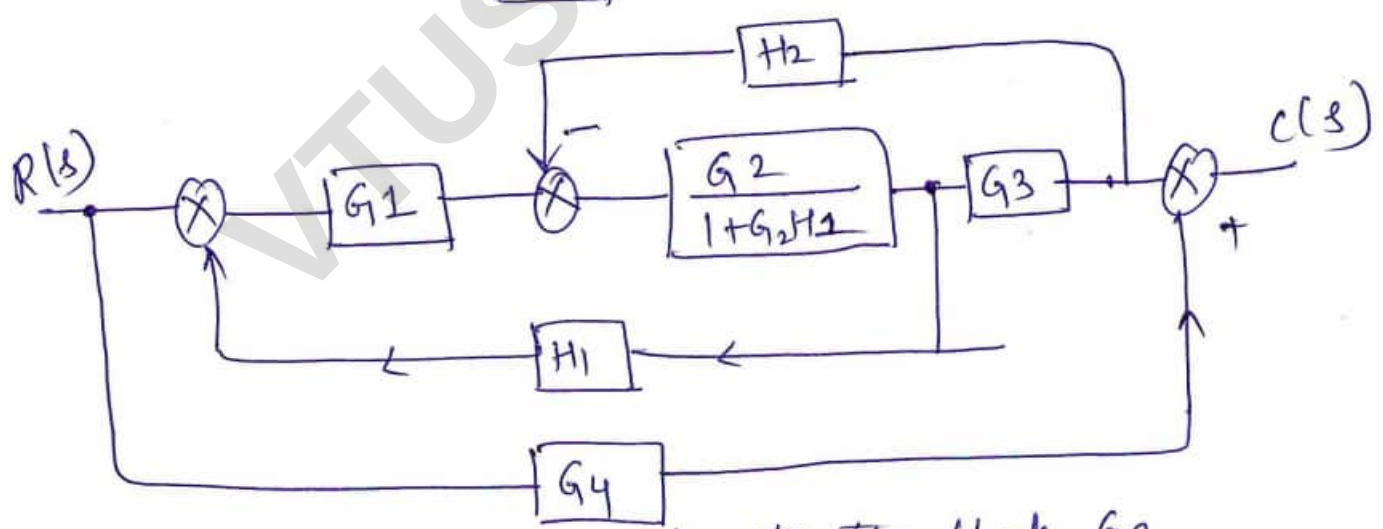
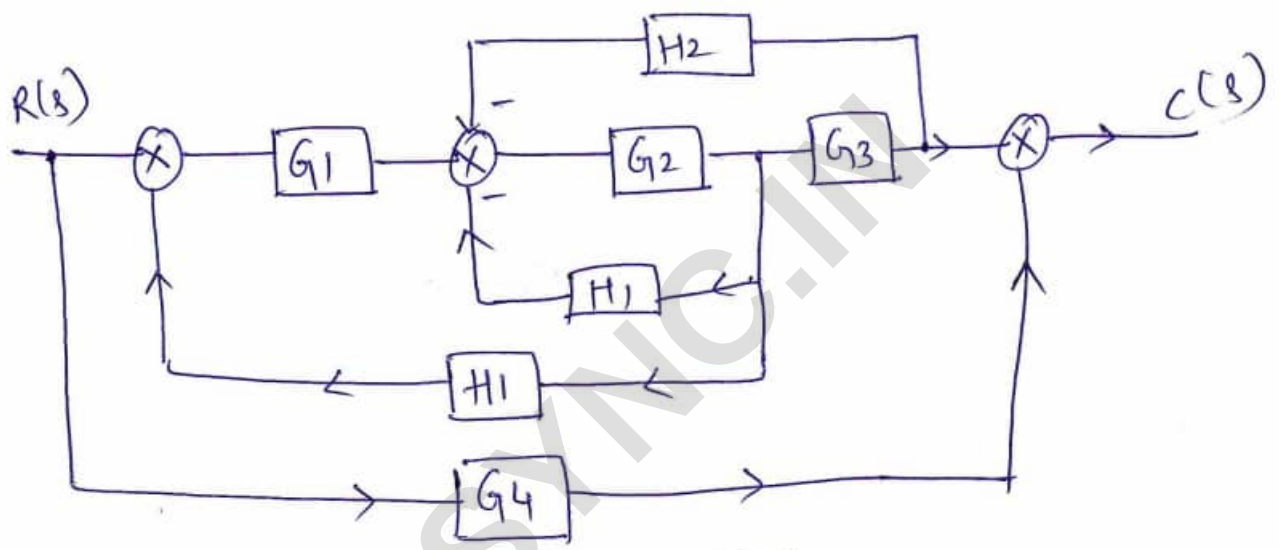
$$\begin{aligned}
 & \frac{G_2(1+G_3)}{1+G_2} \\
 &= \frac{1 + \frac{G_2(1+G_3)}{1+G_2} \cdot \frac{H_1(H_2 + H_2G_3 + 1)}{(1+G_3)}}{1 + \frac{G_2(1+G_3)}{1+G_2} \cdot \frac{H_1(H_2 + H_2G_3 + 1)}{(1+G_3)}} \\
 &= \frac{G_2(1+G_3)}{1+G_2 + G_2H_1 + G_2H_1H_2 + G_2G_3H_1H_2}
 \end{aligned}$$



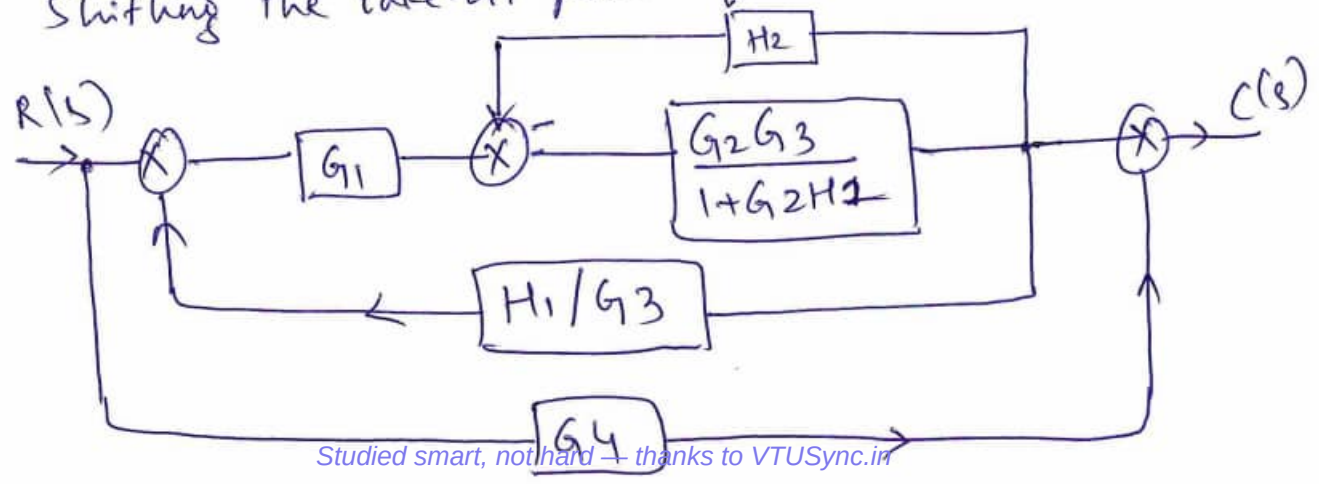
⑦ Reduce the following block diagram :

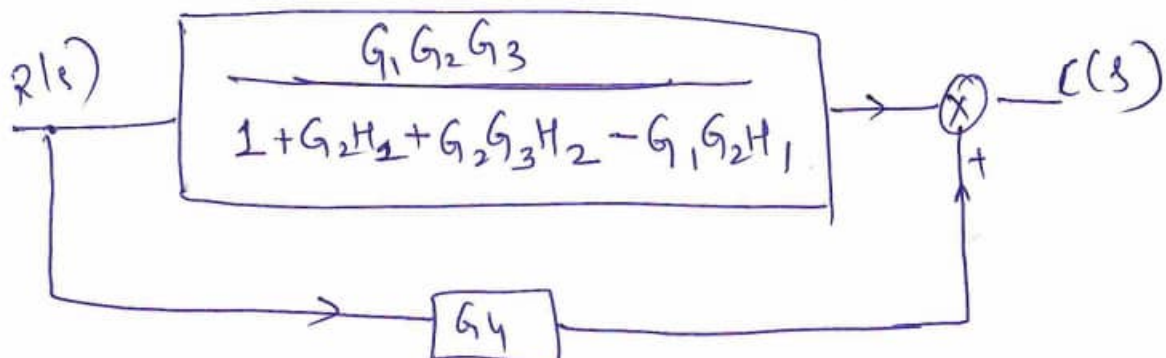
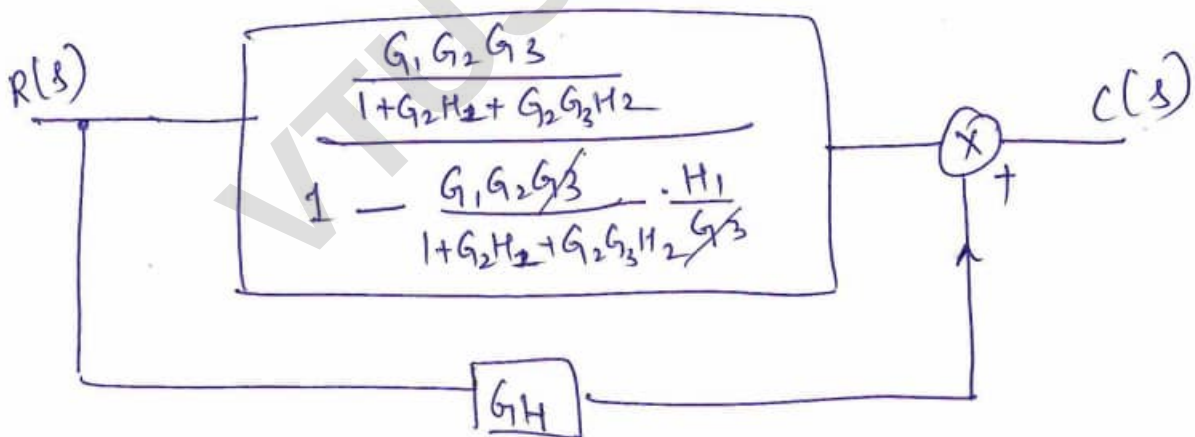
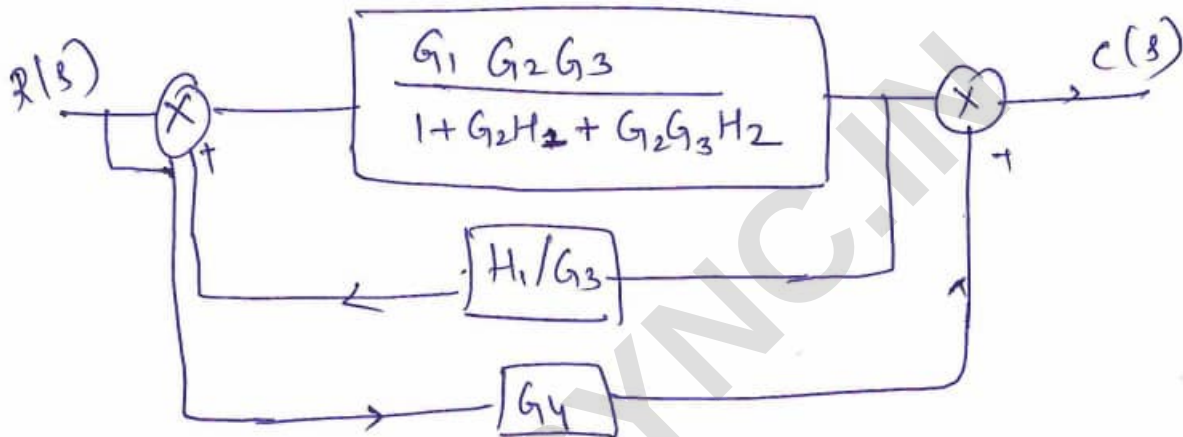
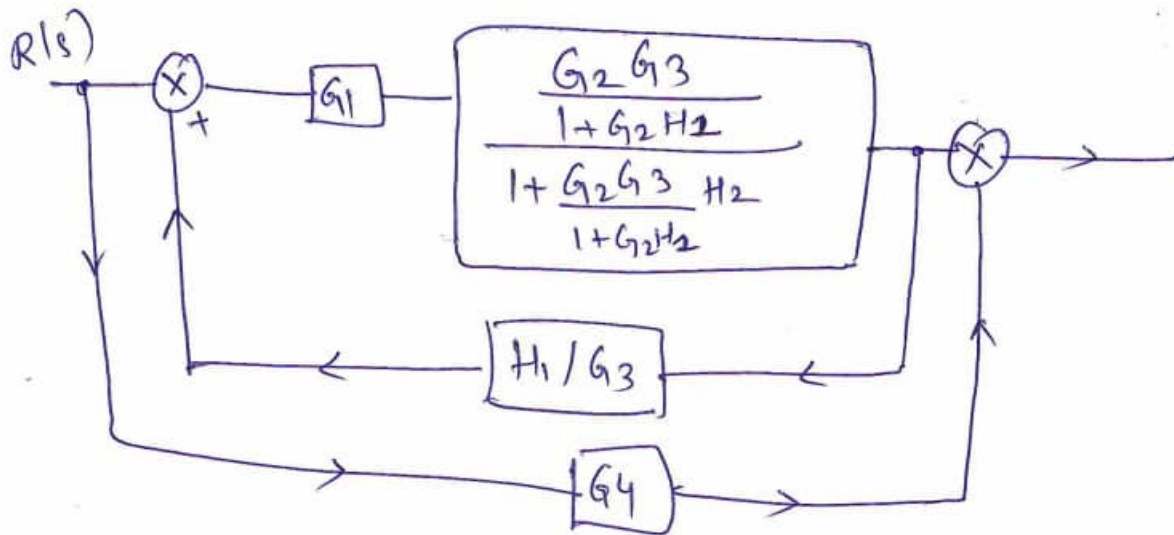


Sol<sup>n</sup>:



Shifting the take-off point after the block  $G_3$







$$R(s) \rightarrow \boxed{G_4 + \frac{G_1 G_2 G_3}{1 + G_2 H_2 + G_2 G_3 H_2 - G_1 G_2 H_1}} \rightarrow C(s)$$

$$R(s) \rightarrow \boxed{\frac{G_4 + G_2 G_4 H_1 + G_2 G_3 G_4 H_2 - G_1 G_2 G_4 H_1 + G_1 G_2 G_3}{1 + G_2 H_2 + G_2 G_3 H_2 - G_1 G_2 H_1}} \rightarrow C(s)$$

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